

MOS Devices and Integrated Circuits for Ionizing Radiation Dosimetry: a Review

Invited Paper

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Abstract—In this paper we review the main aspects of MOS dosimetry. MOS dosimeters have been used for the last 50 years, and its performance has improved since then. To understand how a MOS dosimeter works its limitations and the different strategies to overcome them, it is important to overview the radiation effects in the MOS structure and how it can be used for dose estimation. Different reading methods are presented that improve some aspects of the performance of the MOS dosimeter, such as its lifespan or the temperature dependence of the dosimetric signal. Different well established MOS devices used for dosimetry are reviewed, pointing out its advantages and disadvantages. We finish with the description of some integrated circuits specially designed with the aim of outperform those MOS devices and incorporate more functionality to the dosimeter.

Index Terms—MOS Dosimetry, Ionizing Radiation, Radiotherapy, MOSFET, Floating Gate Transistor, Solid-State Detectors.

I. INTRODUCTION

THE use of MOS devices to estimate ionizing radiation absorbed dose has been proposed in 1970 by W. G. Poch and A. G. Holmes-Siedle [1], and since then, a considerable amount of related research has been carried out and published. The MOS dosimeter was first presented as a radiation sensor for space applications [2], but the evolution and improvement of MOS dosimetry extended the application to more challenging fields such as modern radiation oncology modalities [3]. In the quest of improvement, not only have specially designed MOSFETs been used as dosimeters, but also many Application Specific Integrated Circuits have been designed and characterized as dosimeters. The aim of this paper is to give an overview of the advances in MOS dosimetry over the years. We will introduce the reader to the physical mechanisms involved in MOS dosimetry, and we will discuss their main advantages and limitations, and how some of these limitations were overcome with different reading techniques and integrated circuit design.

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A. Applications

The main applications of MOS dosimetry are radiotherapy dose control and space radiation monitoring. Regarding the medical field, MOS dosimeters have several advantages in comparison with other frequently used dosimeters as TLD or diodes. They are easy to read since the dosimetric signal is a voltage that has cumulative response with dose. Thus, they can be used both in passive mode (reading before and after the exposure) and in real time measurements. They have excellent surface response, are dose rate independent, and have a very small size. International medical and radioprotection organizations [4], [5] recommend the use of in-vivo dosimetry in radiotherapy for patient safety and quality assurance. Measurements should be independent from the radiotherapy system for a reliable control of the delivered dose. The required accuracy is 3% [4] in every single irradiation session to achieve a 5% in the whole treatment. France, Norway, Sweden, Finland, Austria, Denmark and Czech Republic, already included in-vivo dosimetry as a mandatory practice [6]. Belgium and the United Kingdom declared it a suggested practice. It is expected that other countries join this regulations in the near future.

As well, the radiology control during intervention is a concern, not only for patients, but also for medical and assistant personnel.

Regarding space applications MOS dosimetry has the advantage of the direct measuring of total ionizing dose (TID) damage on CMOS integrated circuits [7], [8], [9]. Finally it can be mentioned that MOS devices are suitable for high energy physics instrumentation [10], [11].

B. Physical working principle

When an ionizing particle interacts with matter electron-hole pairs are created along its path. The ionizing energy absorbed per unit of mass is called dose. In the international system it has units of gray, $\text{Gy} = \text{J/kg}$. Another frequently used unit is the rad which is 0.01 Gy . Considering MOS dosimetry we will focus our description on the gate oxide of the structure. The energy required to create a single pair in SiO_2 is $17 \pm 1 \text{ eV}$ [12]. From this value it can be calculated the

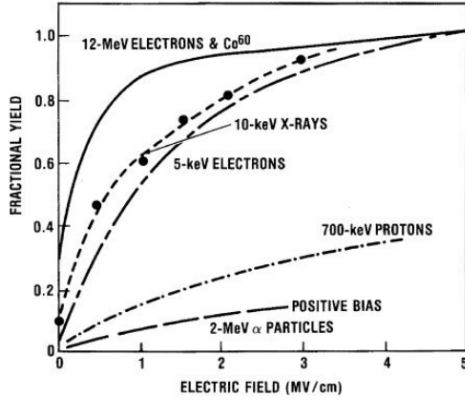


Fig. 1. Fractional charge yield for different types of particles [13].

e-h density created per unit of dose, $g_0 = 8.1 \times 10^{12}$ pairs/rad. After their creation a fraction of these pairs suffer instant recombination. In the presence of an electric field electrons and holes are separated reducing the initial recombination thus increasing the fractional charge yield. The fraction of pairs escaping recombination is determined by the magnitude of the electric field and the initial line distribution of the generated charge. The line distribution depends on the radiation-matter interaction mechanisms which are related to the type of incident particle and its energy [13]. Fig. 1 shows the fractional yield dependence with the applied electric field for different particles.

With a positive voltage applied to the gate, the electric field within the oxide drifts electrons towards the gate and holes in opposite direction to the Si-SiO₂ interface. Electrons are swept out the oxide rapidly, but holes start a slow and dispersive transport [14]. In their path to the silicon substrate they can get trapped within the oxide. It is usually assumed that the traps are oxygen vacancies associated with an incomplete oxidation process. Trapped holes are relatively stable in long term and behave as net positive charge located in the insulator. Besides, hole transport can activate an hydrogen related process which releases a proton that is also drifted by the electric field. When it arrives to Si-SiO₂ interface it can react with other hydrogen atom previously bonded to a Si atom. The generated hydrogen molecule diffuses out of the structure leaving a free Si bond at the interface. This dangling bond is an active electrical state in the forbidden energy gap of silicon and is usually known as interface trap [15]. Charge trapping and its sign depends on the energy of the trap and the Fermi level at the interface.

The positive charge buildup and the interface traps creation have electrical consequences on the MOS structure. Both contribute to the shift of the threshold voltage (V_T):

$$\Delta V_T = -\frac{1}{\epsilon_{SiO_2}} \int_0^{t_{OX}} \rho(x) x dx - \frac{\Delta Q_{IT}}{C_{OX}} \quad (1)$$

where x is the distance from the gate towards the substrate, t_{OX} is the gate oxide thickness, $\rho(x)$ is the volumetric density of trapped charge, Q_{IT} is the superficial density of charge

due to interface traps and ϵ_{SiO_2} is dielectric permittivity of the silicon dioxide. Analyzing Eq. 1 it can be concluded that the contribution to the shift of V_T due to oxide traps (ΔV_{OT}) is negative since holes are positive charged, and that charge trapped close to the interface has more impact in the shift of V_T than that trapped close to the gate. Regarding the contribution of interface traps (ΔV_{IT}), it was observed that trapped charge sign depends on the Fermi level at the interface. For p-channel transistors in inversion the net charge at the interface is positive, but for n-channel transistors the net charge contribution is negative. Interface traps also affects the subthreshold characteristics of MOSFETs by reducing the slope of the I-V curve [16].

In a simplified model, the oxide traps can be considered close to the interface. Radiation-induced holes are created across the insulator and drifted toward the substrate by the electric field. Then, to estimate oxide traps charge we can use

$$\Delta Q_{OT} = \int J_h \cdot f_T(E) dt \quad (2)$$

being f_T the fraction of holes trapped in the oxide which is a function of the electric field E , and J_h is the flux of holes arriving traps close to the interface. Its expression can be written as

$$J_h = q \cdot g_0 \cdot t_{OX} \cdot f_Y(E) \cdot \frac{dD}{dt} \quad (3)$$

where f_Y is the fractional charge yield, $\frac{dD}{dt}$ is the dose rate and $g_0 \cdot t_{OX}$ accounts for the creation of radiation-induced pairs per unit area along the volume of the gate oxide. From Eqs. 2 and 3 we can obtain the absorbed dose dependence of the contribution of the oxide trapped charge to the shift of the threshold voltage as

$$\Delta V_{OT} = -\frac{t_{OX}^2}{\epsilon_{OX}} \cdot q \cdot g_0 \cdot f_Y(E) \cdot f_T(E) \cdot \Delta D \quad (4)$$

In a more realistic model, it should be taken into account that the charge trapping process continuously modifies the local field inside the oxide changing in turn the generation yield, transport and capture rate itself. Therefore the continuity equations for both type of carriers, capture, and Poisson equations must be solved simultaneously in an auto-consistent way. This approach yielded remarkable results in reproducing the evolution of V_T after bias changes during irradiation [17], [18].

C. Radiation sensitivity

MOS dosimeters are built with thick gate oxide (high t_{OX}) to improve the ΔV_{OT} shift with dose. Thus, neglecting ΔV_{IT} we can consider $\Delta V_T \approx \Delta V_{OT}$ and the radiation sensitivity can be defined as:

$$S_R = \frac{\Delta V_T}{\Delta D} \approx -\frac{t_{OX}^2}{\epsilon_{OX}} \cdot q \cdot g_0 \cdot f_Y(E) \cdot f_T(E) \quad (5)$$

The quadratic dependence of the sensitivity with the gate oxide thickness was experimentally verified in [19].

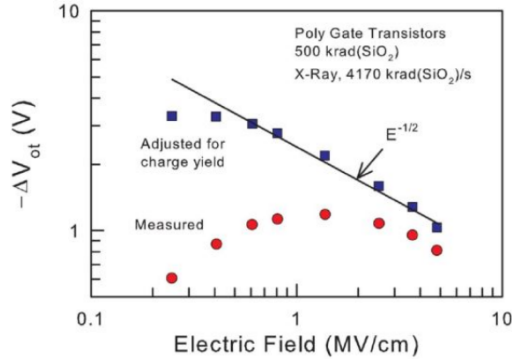


Fig. 2. Electric field dependence of ΔV_{OT} . Measured data shows a maximum value close to 1 MV/cm. Adjusted data with fractional charge yield shows the capture cross section depends with $E^{-0.5}$ [20].

Applying a gate voltage during the irradiation allows to modulate the electric field in the oxide which influences in the fractional charge yield and the capture probability. Fig. 2 shows the sensitivity vs the oxide electric field. For low fields, the sensitivity is small due to the low fractional yield. Sensitivity grows as the field increases reaching a maximum value. For higher fields the sensitivity falls due to the reduction of the hole capture cross section [20].

The interested readers can consult several review papers for a deeper description of the radiation effects in MOS devices [13], [21], [22], [23].

The sensitivity of the MOS dosimeter can be calibrated by experimentally measure the shift in the threshold voltage ΔV_T for a known applied dose ΔD . This calculated sensitivity can further be used to estimate absorbed dose attending the limitations described below.

D. Limitations

Some considerations must be taken into account when using ΔV_T as dosimetric signal: saturation of the response with accumulated dose; V_T dependence with temperature; fading; dispersion in the response among dosimeters; increase of electrical noise with accumulated dose; radiation beam angle dependence; and photon energy response dependence.

The saturation effect (reduction of the sensitivity) with the increase of absorbed dose is mainly due to the influence of the trapped charge on the oxide electric field [24]. Positive charges weakens the local field reducing the fractional yield, and hence bringing down the charge capture rate.

The shift of V_T due to temperature changes when reading the dosimeter can lead to a mistaken dose estimation. Temperature coefficients in the order of -1 mV/°C can easily produce errors beyond tolerated in medical and space applications [25]. If the temperature is not controlled during the measurement, a reading method that mitigates temperature effects it must applied to reduce the measurement error.

Oxide trapped charge is responsible for the radiation induced shift in V_T may be neutralized after irradiation. This charge neutralization causes a recovery in the threshold voltage, which represents a loss of information in the dosimeter.

This effect is called *fading*. It is known that traps very close to the interface, namely border traps, can exchange tunnel electrons from the substrate [26]. Also, a thermal neutralization takes place for energy traps close to the valence band of the Si-SiO₂. Both neutralization phenomena occur at a rate linear with logarithmic time, but while the tunneling front neutralization rate is dependent with the voltage applied to the gate of the MOSFET after irradiation, the thermal emission front is strongly temperature dependent. Both processes produces the recovery of V_T after the irradiation with a rate that depends on operation conditions, such as temperature, electric field at the interface; and fabrication conditions such as traps distribution and energy.

Dispersion in the response of the dosimeters was observed due to variations in fabrication process. Reports for ad-hoc process showed standard deviation of 5% observed even for devices fabricated in the same wafer [27]. Thicker oxides have reported even higher dispersion reaching 16% for 400 nm MOSFETs [28]. In these cases it is required to calibrate each dosimeter separately, which impacts in the production cost. Another approach is to use standard CMOS processes to reduce the fabrication dispersion. This allows the implementation of a batch calibration to obtain a unique sensitivity [29].

Electrical noise is the ultimate limitation for dose resolution. Taking the noise level of the dosimetric signal as N_{RMS} , we can define the Noise Equivalent Dose (NED) as the ratio N_{RMS}/S_R which is a figure of merit of the noise performance of the dosimeter. In MOSFET devices it was observed that absorbed dose increases flicker noise [30]. This effect is supposed to be originated with the exchange of charge between traps and the conduction channel. The increment of interface traps and the capture of positive charge in border traps due to radiation exposure stimulates this interchange increasing the flicker noise level.

MOSFETs fabricated in a planar technology are not symmetrical devices. The angle of incidence of the radiation beam influences in the response because the path line of the radiation particles within the material differs with angle. Secondary particles, generated by the interaction of photons with matter, contribute to the absorbed dose in the gate oxide, which is reached in different amounts depending on the angle. The packaging material and its shape is also a concern regarding the angular dependence.

For photon energies below 1.25 MeV MOSFETs require special calibration of sensitivity due to the attenuation or the dose enhancement effect that can produce the packaging [31]. Low Z materials are recommended to reduce these effects. For energies in the range of medical accelerators 6-18 MeV MOSFETs are dose rate independent, being excellent for surface measurements and depth profile. In the case of bare MOSFETs, measurements obtained are tissue equivalent [31].

E. Reading methods

The most simple reading method of the MOSFET dosimeter is to measure V_T before and after the exposure with its terminals shortcut. The performance of the device can be

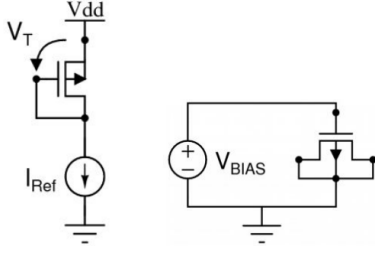


Fig. 3. Left: reading mode. A reference current is forced through the transistor with gate and drain shortcut. The V_{GS} voltage is taken as V_T . Right: biasing mode. The gate is connected to a fixed voltage with other terminals grounded.

improved by biasing the gate with a positive voltage during the irradiation. This is called active mode. Since it requires wires to apply the voltage, real time measurements can be carried out by commuting periodically between biasing mode and reading modes shown in Fig. 3. Time spent in reading mode must be enough to sample V_T , but very short in comparison with time in biasing mode to avoid impact in the response.

The saturation in the response of the dosimeters limits their lifespan. A method to reset V_T to its original value after irradiation was developed in [32], [33] with the aim of re-utilization. The technique consists in the electrical erasure of the trapped charge by means of the injection of Fowler Nordheim tunnel electrons from the gate to the substrate. In their path, these electrons can neutralize trapped holes within the oxide with the resulting recovery of V_T .

Another method, called Bias Cycled Controlled Measurement (BCCM) allows to extend the measurement range of a MOSFET dosimeter by alternating stages of positive and negative biasing [34], [35]. The application of this technique requires real time tracking of V_T . During the positive bias stage, positive charge build up occurs and V_T reduces. When the gate is negative biased, an effect called Radiation Induced Charge Neutralization (RICN) [36] governs the response of the device. The negative biasing inverts the direction of the electric field within the oxide drifting the radiation-generated electrons towards the substrate. In their path they can neutralize a trapped hole, with the consequent recovery of V_T . With the BCCM technique V_T can be bounded between a maximum and a minimum value avoiding the saturation region, and thus, extending the measurement range.

II. MOS DEVICES

A. MOSFET

The use of MOS structures as ionizing radiation sensors was first proposed by Poch and Holmes-Siedle in 1970 [1] and was further described and discussed and finally implemented by Holmes-Siedle in [37]. It was already known that ionizing radiation affects the V_T of MOS transistors, and the proposal was to track this shift in V_T with dose, using the transistor in a "diode" configuration (i.e. gate and drain shorted) and a simple readout circuit to bias the transistor and measure the evolution in the drain-source voltage. The MOS transistor proposed was a p-channel type, with a specially grown gate

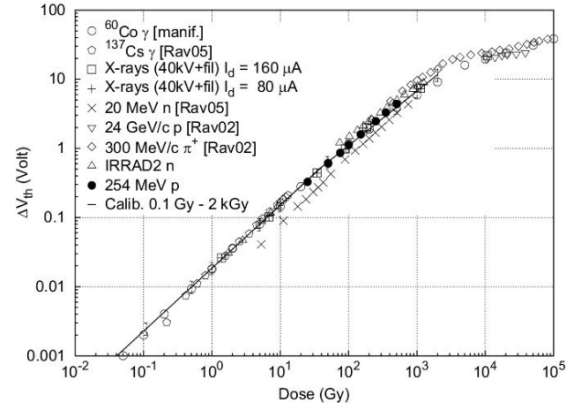


Fig. 4. Radiation response of a RADFET (REM-501C Type K, $t_{ox} = 0.25 \mu\text{m}$) to different radiation beams with zero gate bias. The reference current is $I_{REF} = 160 \mu\text{A}$ [44].

oxide in order to enhance radiation sensitivity: the time and temperature of the oxide growth process was modified to obtain a thick gate oxide and the post oxidation annealing was changed to have a large density of hole-traps. Even though this was an early attempt to use a MOS transistor as a radiation sensor, several aspects of its performance were discussed - sensitivity, linearity, response saturation, temperature effects-, as well as optimization strategies to improve performance - oxide thickness, trap density. The RADFET, as it was baptized later in [38] "The RADFET system for real-time dosimeter in nuclear facility", was proposed for several application fields, but it was first used as a dosimeter in space environment [2]. Since then, the fabrication process of the RADFET was modified in order to improve the dosimetric response, and results of characterization of different batches under different test beams were reported throughout the years [39], [40], [41], [42], [43]. Fig. 4 shows the radiation characterization of the RADFET type K.

Besides space environment, an increasing application for MOSFET dosimeters is radiotherapy and radiology dosimetry. Since mid 90's, research in this field has increased [45], [46], [47], [48], [3] and the main drawbacks addressed are accuracy, angular response, energy dependence and the depth dose profile [3]. The MOSFET dosimeter has the main advantage of a very small dosimetric volume that makes them suitable for radiotherapy applications, where high dose gradients may be found [3], [49]. However, the materials that comprise the packaging surrounding the sensitive device affect the dosimetric response of MOSFET dosimeters [31]. In particular, the thickness of the packaging makes it difficult to follow the recommendations of the International Commission on Radiological Protection (ICRP) to measure an effective depth dose at 0.07 mm [50], known as skin dose. In this sense, Kwan et al. [49] proposed a specially designed MOSFET dosimeter, whose packaging technology provides a Water Equivalent Depth (WED) compatible with the ICRP recommendations, 0.07 mm. The gate oxide of the MOSkin is 550 nm, and the chip is mounted below the surface of the carrier, as shown in

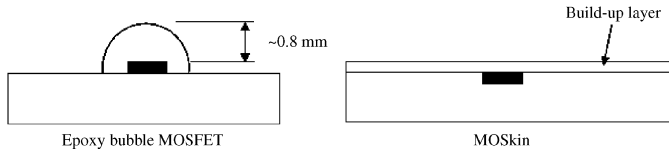


Fig. 5. Comparison between a usual RADFET packaging (left) where an epoxy bubble covers the die with a 0.8 mm thickness, and the MOSkin packaging (right), where the die is below the surface of the carrier, and it is covered with a 0.07 mm build-up layer for skin dose measurements [49].

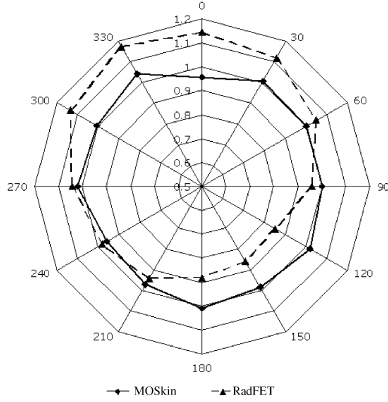


Fig. 6. Angular response of the MOSkin and RadFET detectors when exposed to an ^{192}Ir source within a solid water phantom. [49].

Fig. 5, where an schematic comparison with the RADFET is made. Despite the fact that the MOSkin is not geometrically symmetric, its angular response is much better than that of the RADFET, Fig. 6, due to the extremely thin packaging. Since then, the MOSkin was thoroughly calibrated [51], [52] and used in several radiotherapy applications, such as prostate brachytherapy [53], gynecological brachytherapy [54].

The RADFET and the MOSkin are specially fabricated dosimeters that need an ad-hoc fabrication process in order to grow thick gate oxides for the high responsivity. The technology trends make the gate oxide from standard CMOS fabrication process to be thinner as time goes by. Therefore, it is not possible to fabricate a highly sensitive dosimeter in a commercial process using standard devices. A procedure to overcome this limitation was proposed through the use of the Field Oxide passivation insulator as the gate oxide for a high sensitive MOS dosimeter, the FOXFET [55]. It was already known that radiation also affects the Field Oxide which can affect the leakage current in standard devices [56]. As field oxides are much thicker than standard gate oxides, a device where the gate oxide is replaced with the field oxide, as shown in the cross-section of the FOXFET in Fig. 7, makes a much more sensitive dosimeter. In [55] FOXFETs fabricated in two different commercial CMOS processes were characterized, and they showed a behaviour comparable to the RADFET. The advantage of the FOXFET is that it can be fabricated in the same silicon die as peripheral circuitry in any CMOS process. This characteristic was later exploited in different circuits that will be discussed in the next section

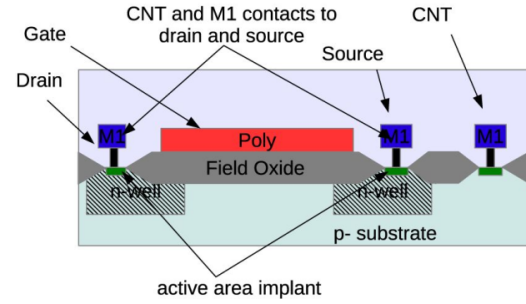


Fig. 7. Cross-section of the FOXFET Dosimeter. The Field Oxide is used instead of the standard Gate Oxide to increase sensitivity [55].

[57]. The device was characterized for in vivo dosimetry in radiotherapy [58] showing suitability for this application. The depth dose profile is in good agreement with what is expected from measurements with a ionizing chamber. The angular response measured, is worse than the MOSkin's but better than the RADFET's, and it is suggested that it can be improved if special care is taken in the packaging design.

One of the issues that must be addressed with MOSFET dosimeters is the effect of temperature on the threshold voltage that can lead to erroneous dose estimations. The RADFET, the MOSkin and the FOXFET report dependence in the dosimetric signal with temperature [39], [59], [60]. All MOS transistors exhibit a temperature sensitivity, and there have been several proposals throughout the years in order to mitigate this effect. The three main strategies to reduce the temperature effect on MOS dosimeters are:

- i) Bias the transistor with a drain current where the temperature coefficient is minimum, the I_{MTC} .
- ii) Perform differential measurements with two MOS dosimeters with different radiation sensitivity, but similar temperature sensitivity.
- iii) Characterize the dosimeter response to temperature in order to subtract the temperature contribution to the shift in the dosimetric signal.

We will now focus on item i) and in the following section we will discuss the second strategy. Temperature affects mainly two parameters of the MOS transistor, the mobility (μ) and the threshold voltage. Both parameters decrease with increasing temperature, but they have opposite consequences on the drain current. A lower mobility means a lower drain current, while a lower V_T means a higher current. This opposite contribution yields a current for which the temperature coefficient of the measured gate voltage is minimum: The Minimum Temperature Coefficient Current (I_{MTC}). The I_{MTC} has been measured for several MOS dosimeters [61] [28], [39], [25], [62], [55] and it is a recommended reference current for dose measurements. However, it has been reported that the temperature coefficient is affected by radiation [39] [58]. Fig. 8 shows the change of the thermal coefficient with dose for a RADFET type K. This effect is equivalent to the variation of the I_{MTC} with the absorbed dose of the dosimeter [62]. It was suggested in [62] that the change in I_{MTC} with total dose is

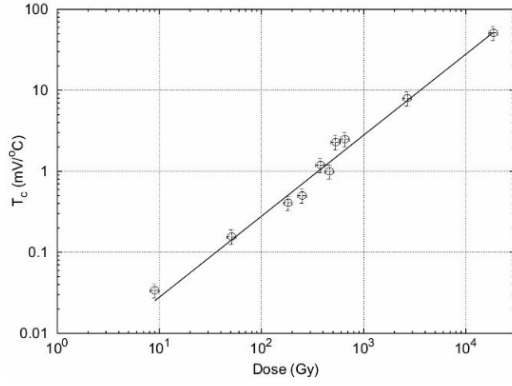


Fig. 8. Shift in the temperature coefficient (T_C) of a RADFET (REM TOT-501C Type K, $t_{ox} = 0.25 \mu\text{m}$) with radiation. The T_C was measured at $I_{REF} = 160 \mu\text{A}$ [39].

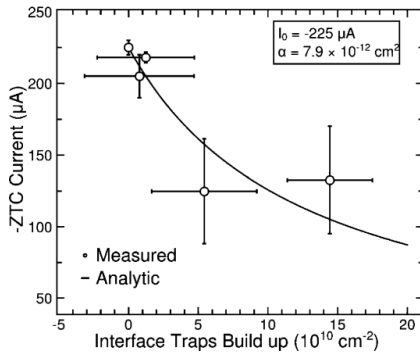


Fig. 9. Correlation between the I_{MTC} (also called I_{ZTC} for zero temperature coefficient) and the depassivation of interface states [62].

correlated with the interface traps depassivation that affects the mobility. Fig. 9 is the plot of I_{MTC} against the increment of radiation-induced interface traps. The variation of I_{MTC} with dose makes this a limited method for temperature mitigation.

B. Floating Gate Dosimeter (FGDOS)

Floating Gate (FG) transistors are MOSFETs where the polysilicon gate is surrounded by insulator and hence it is electrically isolated. As any other MOSFET, it is affected by ionizing radiation, and can be used as a dosimeter. Kassabov et al. [63] have characterized the radiation effects on a FG-MOSFET with a control gate deposited above the floating gate. Before irradiation, the floating gate of the device under test was either negatively or positively charged. They conclude that when the FGMOSFET is negatively charged, radiation induced holes transport towards the FG, neutralizing the electrons, increasing the threshold voltage seen from the control gate. With this result, this is the first time a FGMOSFET is proposed as a dosimeter, with a sensitivity of approximately 10 mV/Gy [63].

Considering the charge neutralization in a Floating Gate, Peters et al. [64] proposed in 1996 a novel design for a floating gate dosimeter: the floating poly-silicon layer deposited over the gate oxide should be extended over field oxide in order to capture more radiation generated charge. A schematic diagram

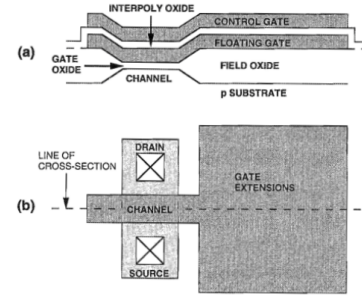


Fig. 10. Cross-section (a) and Layout view (b) of the FGDOS proposed by Peters et al. where the floating gate is extended over Field Oxide [64].

of the cross-section and layout of the device is shown in Fig. 10. The device was fabricated in a 5 μm nMOS non-commercial process, with a resulting gate oxide of 50 nm, a inter-poly oxide of 140 nm, and a field oxide of 1 μm . The control gate allows to bias the dosimeter during irradiation and improve sensitivity, achieving 280 mV/Gy when the control gate bias is 15.5 V.

An improved version of the FGDOS was later presented by Tarr et al. [65]. The main differences with Peters' proposal are the addition of an injector electrode to charge or discharge the floating gate, a smaller extension of the control gate layer to reduce its capacitance, and differential topology. The injection of charge in the FG allows to get rid of the control gate bias during irradiation, as the FG charge itself force an electric field in the oxide that favors the electron-hole transport. If both FG in the dosimeter are charged with different charge, a differential output can be measured. This version of the FGDOS was fabricated in a 1.5 μm commercial CMOS process with 27 nm gate oxide, 48 nm inter-poly oxide, and 600 nm field oxide. The initial sensitivity measured was 70 mV/Gy, with a little loss of sensitivity with subsequent injections for re-use.

In [66], the same research group continues to improve the FGDOS. The new proposal eliminates the control gate completely, as it affects the sensitivity [67]. Moreover, a reference transistor of identical geometry is fabricated along with the FGMOSFET for temperature compensation, and the dosimetric signal is the gate voltage needed to bias the reference transistor with the same drain current as the FGMOSFET. For the implementation of this dosimeter, an operational amplifier is added to the circuit which is responsible for maintaining the same drain current in both the FGMOSFET and the reference MOSFET. The schematic of the circuit and a cross section of the FGMOSFET is shown in Fig. 11.

The approach of Tarr et al. in [66], changes the idea of a single MOSFET as a dosimeter, but instead the MOSFET is part of a more complex circuit in order to produce a signal that changes with the dose. In the next section, we present several circuits used as dosimeters, were the sensitive element is neither one or a pair of MOSFETs, FGMOSFETs or FOXFETs.

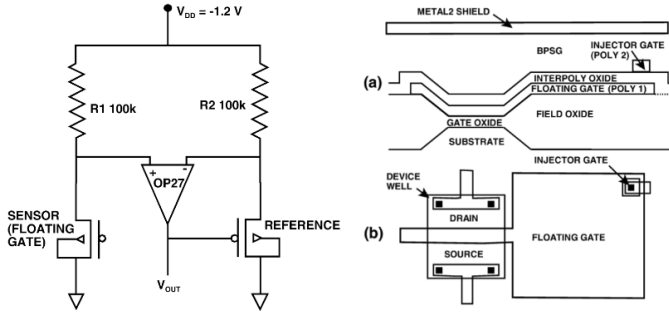


Fig. 11. Cross-section (a) and Layout view (b) of the FGDOS proposed by Tarr et al. as an improvement from [10]. The control gate is eliminated, an injection electrode is added, and a metal 2 shield is also added to mitigate noise. On the left, the reading circuit is presented [66].

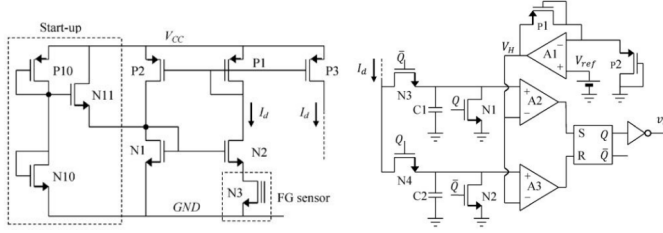


Fig. 12. Left (a) circuit with FG device as the reference to define the current. Right (b) current-frequency conversion circuit [68].

III. CIRCUITS

Several integrated circuits were proposed for ionizing radiation dosimetry. In this section we will present some CMOS implementations which successfully integrate the MOS sensing device and the reading circuit in a single monolithic chip.

The approach developed in [68] is a dosimeter with frequency output fabricated in a standard $0.6 \mu\text{m}$ technology. The sensor is a FG device which is pre-charged allowing to sense radiation even with zero bias. The reading circuit of Fig. 12(a) shows how changes in the charge of the FG are converted to current and Fig. 12(b) is the implementation of the current-frequency converter. The complete dosimeter occupies $450 \times 250 \mu\text{m}^2$. Tested with gamma radiation of Cs-137, the initial radiation sensitivity reported is -803 Hz/rad with a temperature coefficient of $-3.57 \text{ kHz/}^\circ\text{C}$ and a supply voltage dependence of 67 Hz/mV . The initial frequency of the output is 1 MHz . The dose range is 1.5 krad , but it could be reset by a new injection of the FG. In a later paper [69] the design was improved regarding the temperature coefficient and the linearity response by the development of an new analog reading circuit. In this case the circuit output is a current, with a sensitivity of $-11.4 \mu\text{A/krad}$ and a measurement range of 3.6 krad .

Another integrated circuit was presented in [70] and [29]. In this case the aim of the design was to develop an implantable device for real time dosimetry in radiotherapy. The concept includes a FG sensor, the reading circuit and a RF transmitter block in the medical implant communication service (MICS) band. The chip was fabricated in a 180 nm standard technol-

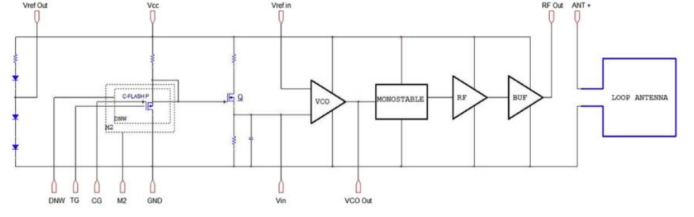


Fig. 13. Integrated circuit schematic proposed in [29] for an implantable medical dosimeter based in a FG sensor.

ogy. The size of the complete system is approximately 1 mm^3 . The FG device is based in a C-Flash cell [71]. Fig. 13 shows a simple analog circuit to read the FG. Its output is the input of a VCO that converts the voltage into frequency in the range of $150\text{-}300 \text{ kHz}$. The circuit also includes a monostable to pulse the operation of output and reduce the power consumption. The pulsed signal is modulated by a RF ring oscillator in the band of MICS which is connected through a buffer to the antenna. The reported radiation test of FG with the reading circuit resulted in a resolution of 1 rad with an accuracy better than 3% over a range of 5 krad . Tests over 2000 devices showed low dispersion in the sensitivity, allowing a batch calibration.

The use of differential sensors was presented in [72]. The main advantages are the temperature effects rejection, the dose measurement range extension and the real time measurement. In [72] two MOSFETs fabricated in the same chip are irradiated with different gate bias applied. In this way each device has different sensitivity and thus different V_T evolution with dose. The dosimetric signal is $V_{T1} - V_{T2}$ which is affected by temperature in common mode and by radiation in differential mode.

Circuits exploiting the differential sensor concept were reported in [60], [73], [74]. All these papers use a pair of FOXFETs as sensor and the rest of the circuits are discrete components. The biasing method used consisted on applying a positive and a negative to the gate of each FOXFET. The negative biased device works under the radiation induced charge neutralization effects and recovers its V_T . Switching the polarity of the biasing between both devices allows to keep the dosimetric signal bounded between a minimum and a maximum value avoiding the saturation in the response. Differential circuits with sensitivity gain are presented and characterized in [73], [74]. The first one shown in Fig. 14 (a) achieves a gain of 91 times (the sensitivity of a single device) but has a poor temperature rejection. It is useful for diagnosis X-ray dose control where the doses are low (below 0.1 rad) and the exposure time is short. The second circuit of Fig. 14 (b) can fix the gain with the ratio between the feedback resistors and has an improved thermal rejection. This circuit is useful for radiotherapy and brachytherapy where temperature changes can occur due to its contact with the patient body, the doses are in the order of 100 rad per session, and exposure times are around 1 minute .

Based on the good results achieved with the proposal

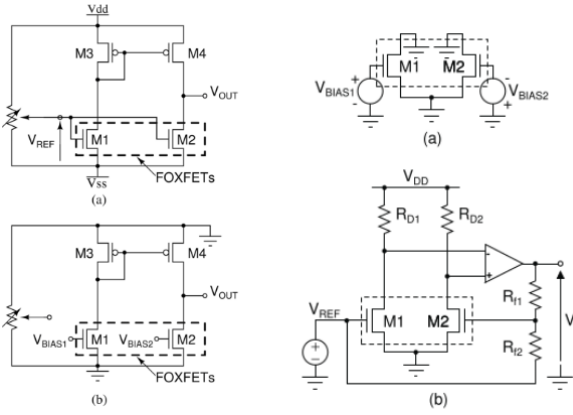


Fig. 14. (Left) Circuit proposed in [73]. (a) Reading mode and (b) Biasing mode. (Right) Circuit presented in [74]. (a) Biasing mode, and (b) Reading mode. Both circuits amplify the radiation sensitivity with differential topology.

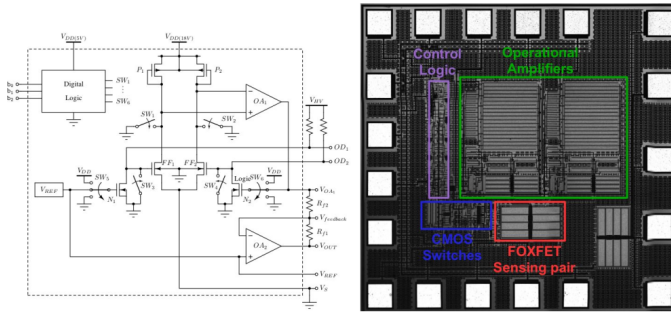


Fig. 15. Schematic of the SoC dosimeter (left) and microphotograph of the chip (right) [57]. The only external components are the resistors whose values allow to set the gain in the sensitivity.

of [74], we addressed its integration in a $0.6 \mu\text{m}$ CMOS technology [57]. Since the sensing pair of FOXFETs had a V_T value close to 28 V a high voltage technology was required. This system on chip (SoC) dosimeter included the switches and logic to bias and measure in differential mode. Fig. 15 shows the schematic and the fabricated chip.

IV. SUMMARY AND CONCLUSION

In this paper we gave an overview of the use of MOS devices for dosimetry. Ionizing radiation effects in the MOS structure were presented to address the physical working principles of the dosimeter. We discussed the main applications and limitations of the use of V_T as the dosimetric signal. Also the different reading methods with their advantages were introduced.

MOSFETs from its origins as radiation sensor and their latter evolution including description of medical applications were detailed. The Floating Gate as an alternative structure for dosimetry was also presented. The main advantage of this device is its fully compatibility with standard CMOS processes.

Integrated circuits exploit the advantages of MOS dosimeters' compatibility with standard CMOS fabrications process. Circuits are focused in the performance improvement of the

dosimetric signal with the inclusion of reading and biasing circuits, or in the addition of special functionalities such as wireless reading.

Further evolution of the MOS dosimeters are expected considering the wide and interesting application areas: medicine, space and high energy physics experiments.

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